

The Structure of Agents

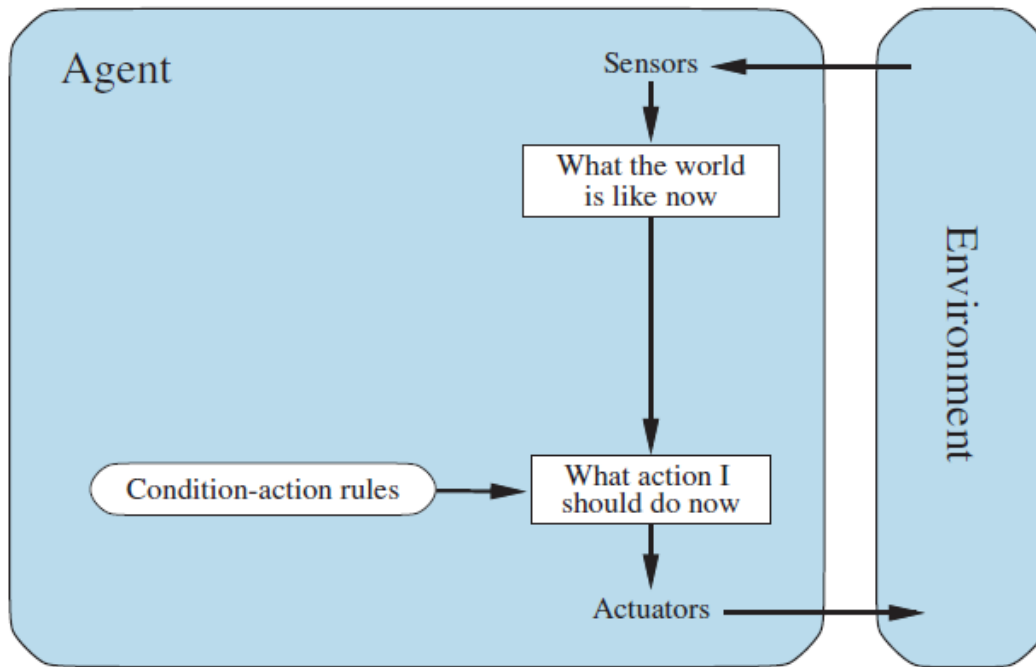
Four basic types embody the principles underlying almost all intelligent systems:

- ◆ Simple reflex agents
- ◆ Model-based reflex agents
- ◆ Goal-based agents
- ◆ Utility-based agents

All of them can be converted into

- ◆ Learning-based agents

Simple Reflex Agent



Rectangles: agent's current internal state
Ovals: background information used in the process.

- Select actions based on the current percepts, and ignore the percept history.

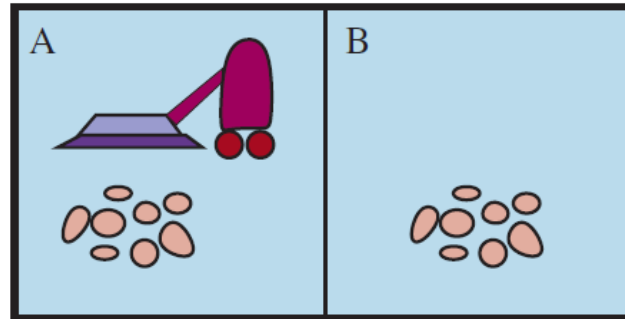
E.g., the vacuum agent

- Implemented through *condition-action* rule.

if dirty *then* suck

if car-in-front-is-braking
then initiate-braking

Vacuum-Cleaner World (Revisited)



if *status* == *Dirty* then return *Suck*
else if *location* == *A* then return *Right*
else if *location* == *B* then return *Left*

Simple Reflex Agent

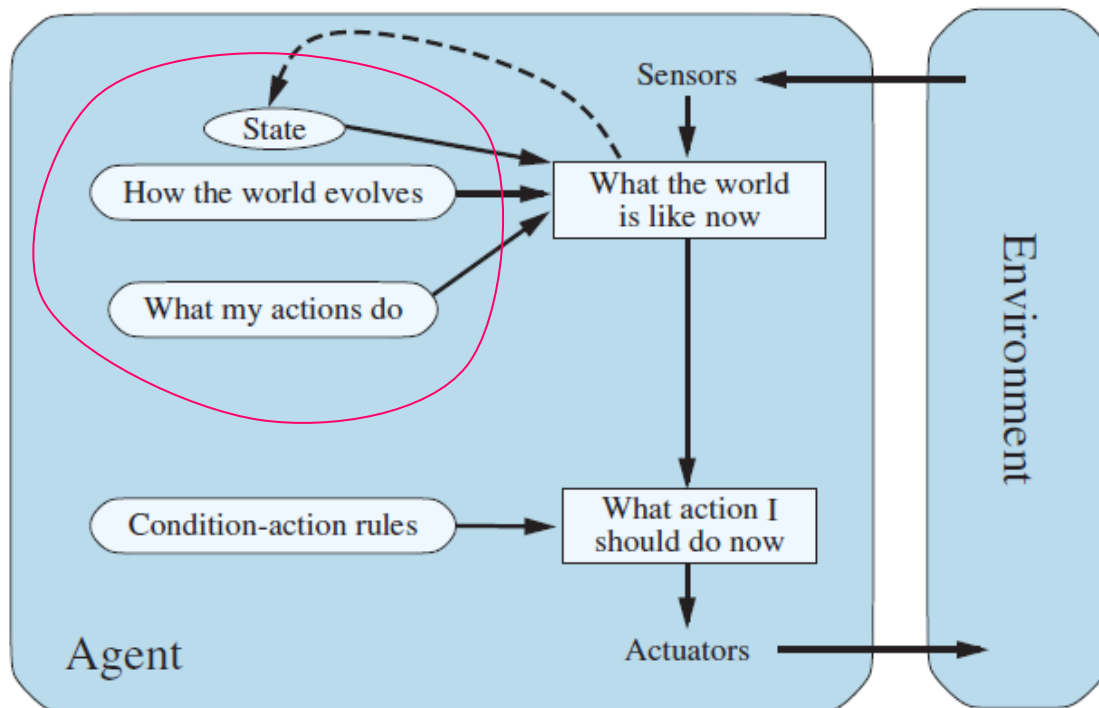
function SIMPLE-REFLEX-AGENT(*percept*) **returns** an action
persistent: *rules*, a set of condition–action rules

state $\leftarrow$ INTERPRET-INPUT(*percept*)
rule $\leftarrow$ RULE-MATCH(*state*, *rules*)
action $\leftarrow$ *rule*.ACTION
return *action*

♣ Limited intelligence

It will work **only if** the correct decision can be made based on only the current percept, i.e., **only if** the environment is fully observable.

Model-Based Reflex Agent



- Partially observable environment.
- Need to maintain some internal state.
- Update it using *knowledge*.



Model of the world

- ◆ How does the world change?
- ◆ How do actions affect the world?

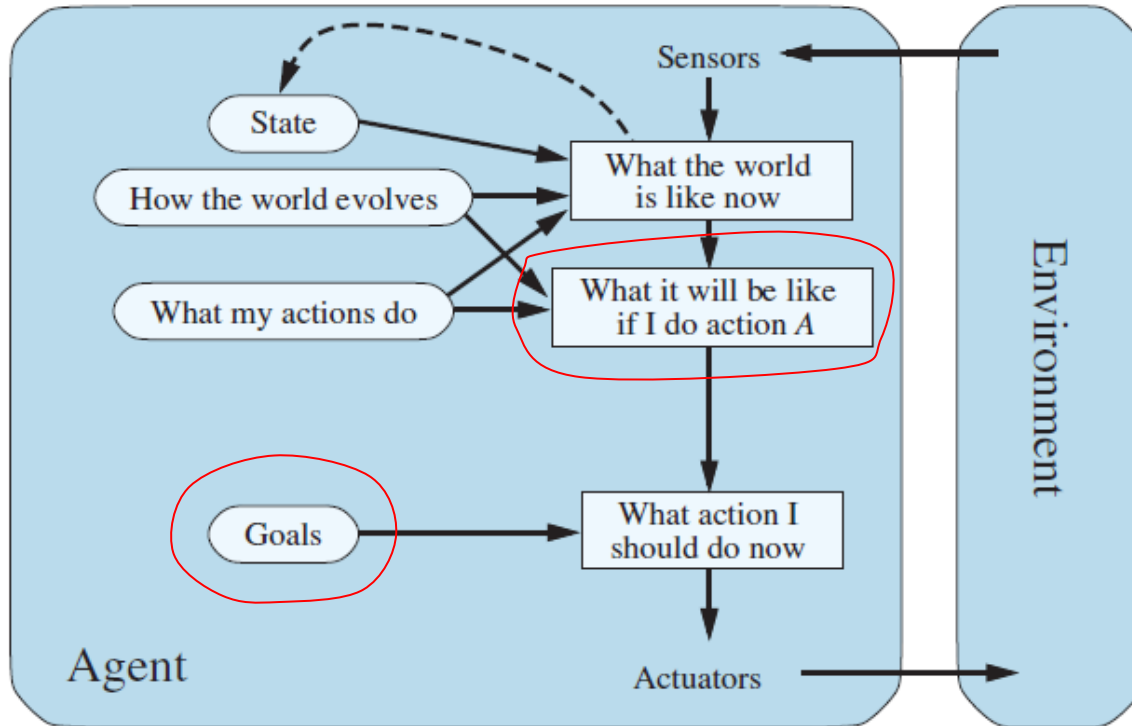
How This Agent Works

function MODEL-BASED-REFLEX-AGENT(*percept*) **returns** an action
 persistent: *state*, the agent's current conception of the world state
 transition_model, a description of how the next state depends on
 the current state and action
 sensor_model, a description of how the current world state is reflected
 in the agent's percepts
 rules, a set of condition–action rules
 action, the most recent action, initially none

state ← UPDATE-STATE(*state*, *action*, *percept*, *transition_model*, *sensor_model*)
rule ← RULE-MATCH(*state*, *rules*)
action ← *rule*.ACTION
return *action*

- It is rarely possible to describe the exact current state of the environment.
- The maintained “**state**” does not have to describe the world.

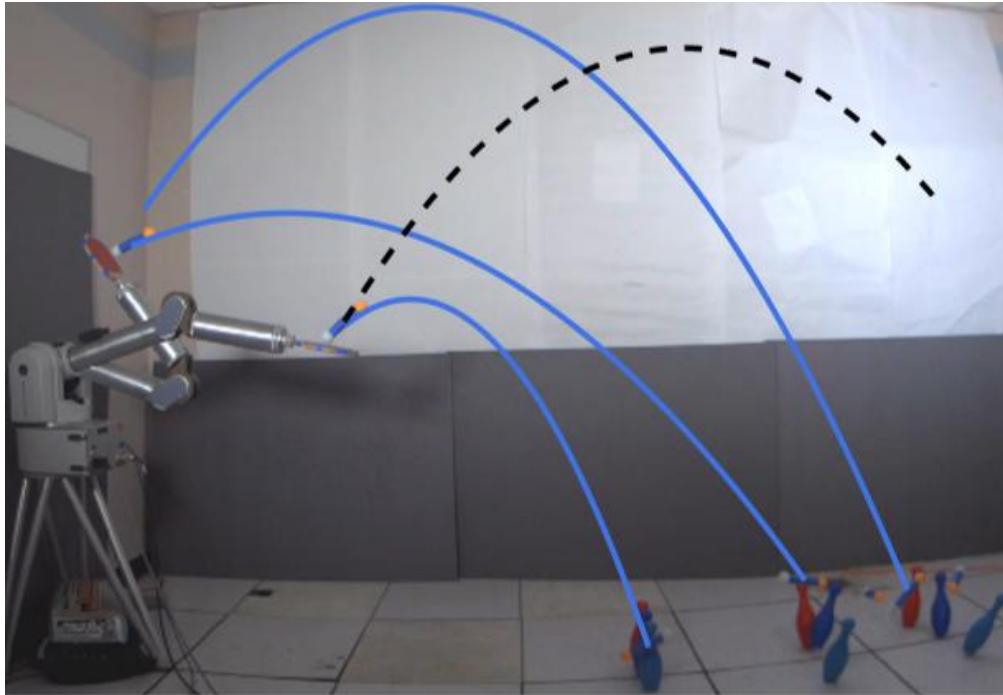
Goal-Based Agent



- Needs also some goal information describing desirable situations.
- *Search* and *planning*
when a long sequence of actions is required to find the goal.
- Difference in *taking the future into account*.

Example: Robotic Batting

Goal Hit an in-flight object to a target.

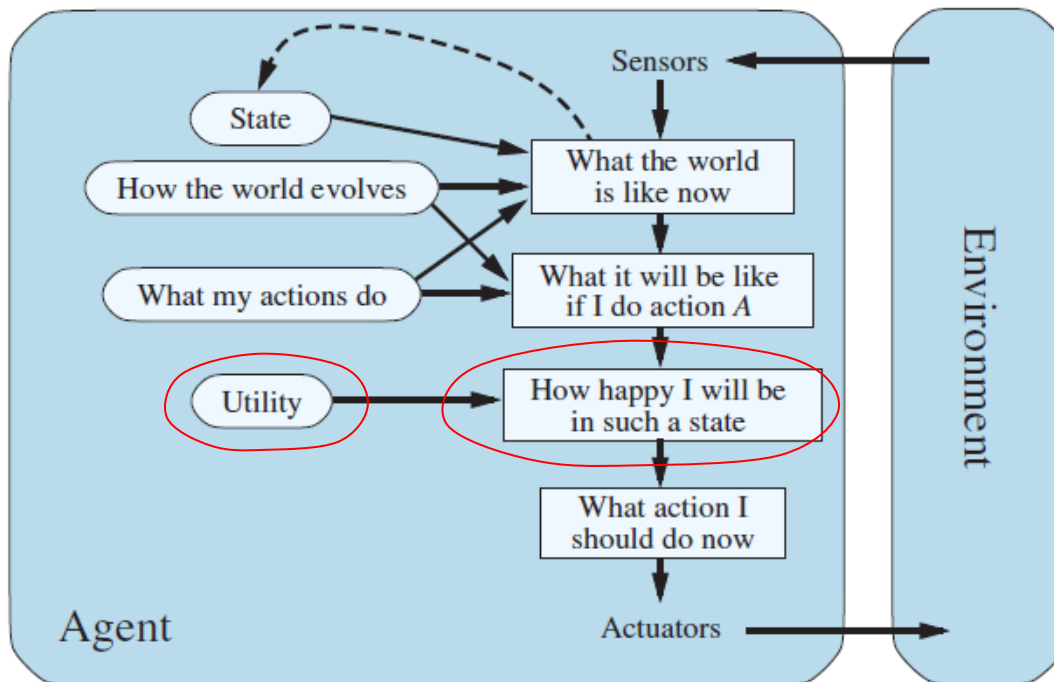


- Sensors: 2 high-speed cameras
- Actuators: 4-DOF WAM Arm
- Models: impact dynamics, aerodynamics & robot kinematics
- State estimation: uses an extended Kalman filter (EKF) to track the velocity & angular velocity of the flying object.
- Planner: determines the time instant of hitting and the bat's position and velocity at this instant.

<https://www.youtube.com/watch?v=dGBevZ54E3s>

[IEEE Transactions on Robotics, vol. 38, no. 5, pp. 3187-3202, 2022](#)

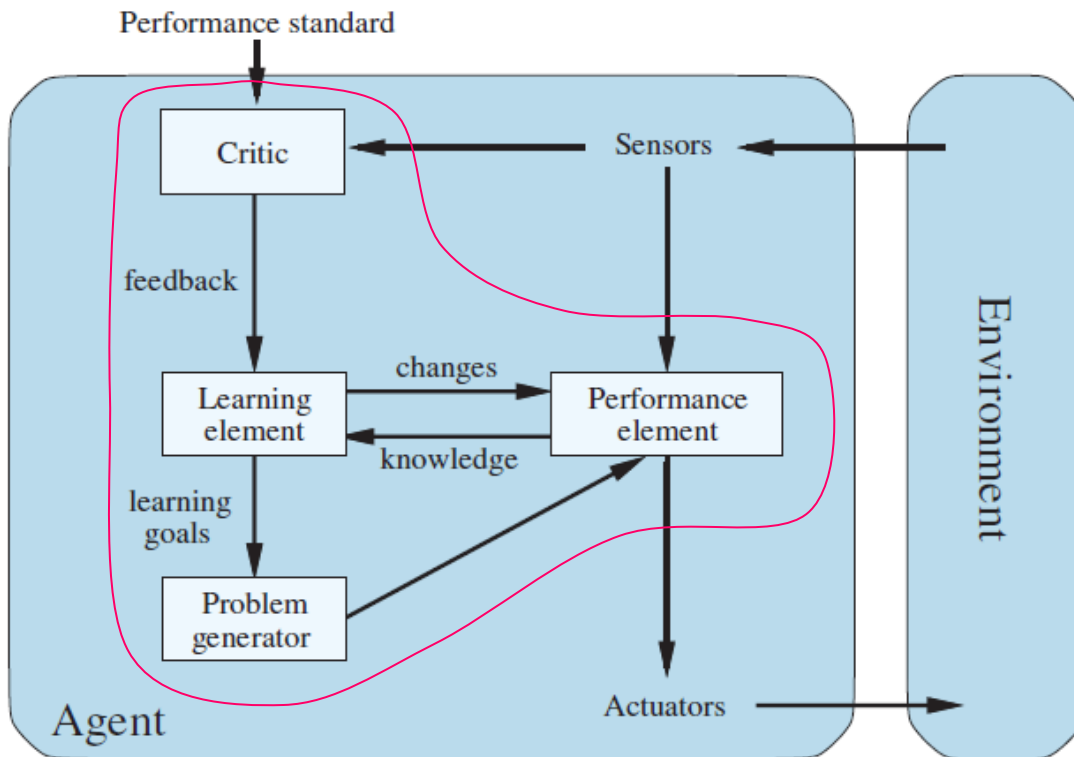
Utility-Based Agent



- Different ways to achieve a goal sometimes.
- Use a *utility function* that maps a (sequence of states) to a real number (*utility*)
↑
internal performance measure
- Maximize expected utility.

- Goal improvements:
 - ◆ selection among conflicting goals
 - ◆ selection based on likelihood of success and goal importance

Learning-Based Agent



- Preferred method for creating state-of-the-art AI systems:

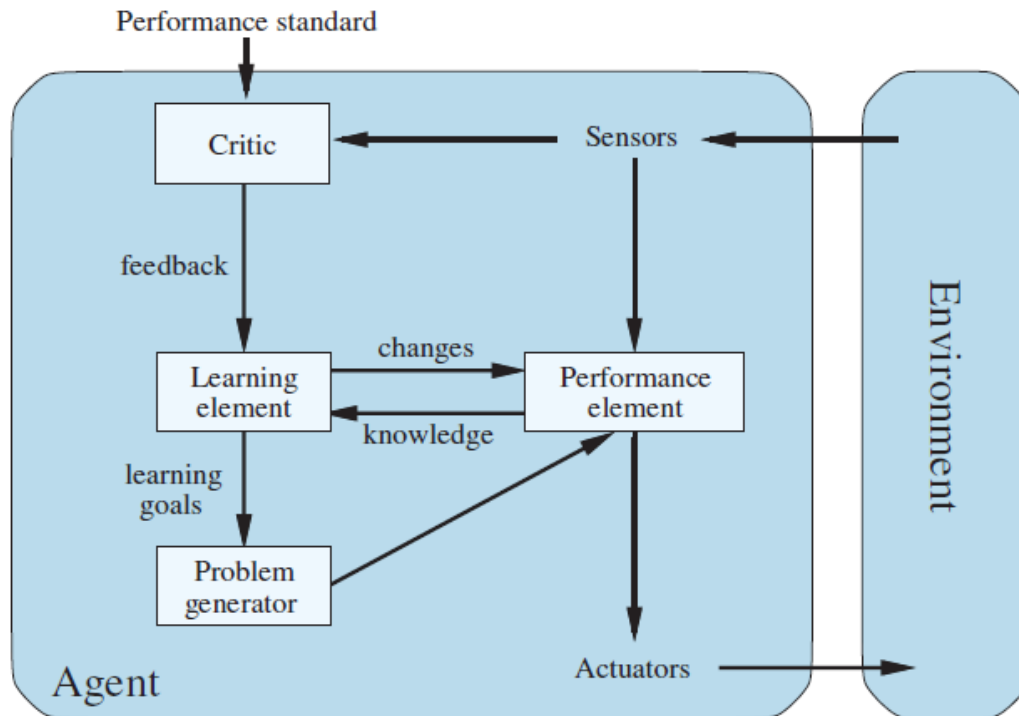
- ◆ Allow operation in initially unknown environments.
- ◆ Adapt to changes in the environment – **robustness**.

- Modifications of the four components to bring them in **closer agreement** with the available feedback



Better overall performance

Learning-Based Agent



- **Critic** provides feedback on the agent's performance based on fixed performance standard.
- **Learning element** introduces improvements in performance element.
- **Performance element** selects actions based on the percepts.
- **Problem generator** suggests actions that will lead to new and informative experiences.